

Computational processing of *Wolbachia*-infected *Drosophila* scRNA-seq libraries

Input: 18 PIPseq libraries from four stocks (two *D. melanogaster* with *wMel*, two *D. simulans* with *wRi*)

Embryos

4 libraries

Primary cell lines

4 libraries, a few weeks in culture

Cell lines

10 libraries (5 lines x 2 replicates)

1 Joint host + *Wolbachia* quantification

kallisto | *bustools* (kb count)

Pseudoalign each library to a combined reference: host transcriptome (*D. melanogaster* or *D. simulans*) + *Wolbachia* genome (*wMel* or *wRi*) + a 16S panel
Per cell: host gene counts, *Wolbachia* gene counts, and 16S counts (to flag other bacteria)

2 Quality control and *Wolbachia* load per cell

scanpy, *scrublet*

Remove low-quality cells and doublets (*Scrublet*, score > 0.4); record UMIs, genes detected, % mitochondrial
Wolbachia load = *Wolbachia* gene UMIs / all UMIs; rRNA-based titer kept for the atlas analyses

3 Common gene space across species

ortholog map

Map *D. simulans* genes to 1:1 *D. melanogaster* orthologs (reciprocal best hits), so every sample uses FlyBase *D. melanogaster* annotation

A. Atlas projection: embryos vs cell lines

13 libraries (embryos + cell lines)

4a Project onto the Flysta3D-v2 embryo atlas

Fixed atlas PCA/UMAP; ingest maps each cell to its atlas neighbours
(no joint re-clustering, so samples cannot shape each other's embedding)
Transfer cell type, tissue, and germ layer with a confidence score

5a Compare composition and similarity

Cell-type composition and Shannon diversity per condition
Pseudobulk Spearman correlations: conditions and atlas tissues
Wolbachia titer by transferred cell type and condition
Tests: exact Mann-Whitney; permutation over all embryo/line relabelings

6a Readouts

Cell lines converge on hemolymph-, gonad-, and CNS-like identities
Cell lines resemble each other more than their own stock's embryos
Embryos, not cell lines, match a single embryonic tissue
Figures 1-4; Tables S9-S11 (results/atlas_stats/)

B. Stage comparison within each stock

4 stocks x 3 stages (embryo, primary cell line, cell line), all 18 libraries

4b Per-stock embedding and pseudobulk

Within-sample HVGs (no ribosomal, mito, heat-shock, stress genes); kNN stage mixing
Pseudobulk counts per library; stocks are the biological replicates

5b Differential expression

PyDESeq2,
~ stock + stage
Depth-matched rerun
GSEA (GO, KEGG)

5c Marker-based cell states

Module scores
per cell
Paired logit t-test
Proliferating flag

5d Polycomb marking

Embryo H3K27me3
S2, BG3 Pc states
Fisher, logistic OR
GO of off genes

5e Primary-culture cells

Rank test vs
cell-line change
Wolbachia load
Species candidates

6b Readouts

Shared stage differences across four stocks and two species
Silenced developmental regulators are Polycomb targets
Proliferating primary cells partly resemble cell lines
Figures 5-9; Tables 1-2, S1-S8, S12-S13 (results/pseudotime/)